

LIST OF EXPERIMENTS

1. **The Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **The Stanford Prison Experiment:** A study on the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **The Asch Conformity Experiment:** A study on the effects of group pressure on individual behavior, conducted by Solomon Asch in the 1950s.
4. **The Pavlov's Dog Experiment:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
5. **The Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.
6. **The Hawthorne Effect:** A study on the effects of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
7. **The Bobo Doll Experiment:** A study on the effects of observing aggressive behavior, conducted by Albert Bandura in the 1960s.
8. **The Bystander Effect:** A study on the diffusion of responsibility in emergency situations, conducted by John Darley and Bibb Latané in the 1960s.
9. **The Heinz Dilemma:** A study on moral development, conducted by Lawrence Kohlberg in the 1950s.
10. **The Blue Eyes/Brown Eyes Experiment:** A study on discrimination and prejudice, conducted by Jane Elliott in 1968.

Note: A minimum of ten experiments from the following should be performed.

1. To study the characteristics of a common-emitter npn or pnp transistor and to find out the values of current and voltage gains.
2. To study the characteristics of a zener diode and find its reverse breakdown voltage.
3. To study the characteristics of a p-n junction diode in forward and reverse bias.
4. To study the input and output characteristics of a n-channel junction field-effect transistor (JFET).
5. To study the characteristics of a light emitting diode (LED).
6. To study the characteristics of a photodiode.
7. To study the characteristics of a solar cell.
8. To study the characteristics of a thermistor.
9. To study the characteristics of a varactor diode.
10. To study the characteristics of a tunnel diode.
11. To study the characteristics of a metal-oxide-semiconductor field-effect transistor (MOSFET).
12. To study the characteristics of a unijunction transistor (UJT).
13. To study the characteristics of a silicon-controlled rectifier (SCR).

14. To study the characteristics of a triac.
15. To study the characteristics of a diac.

(A) Hardware based experiments

Hardware based experiments are experiments that are conducted using physical equipment and tools. These experiments are often used to test the performance and capabilities of hardware systems, such as computer processors, memory, and storage devices. Some examples of hardware based experiments include:

1. Benchmarking: This involves running a series of tests on a hardware system to measure its performance and compare it to other systems.
2. Stress testing: This involves subjecting a hardware system to extreme conditions, such as high temperatures or heavy workloads, to test its reliability and durability.
3. Compatibility testing: This involves testing a hardware system with different software and hardware configurations to ensure that it is compatible and can function properly.
4. Usability testing: This involves testing a hardware system with real users to evaluate its ease of use and user satisfaction.

Hardware based experiments are an important part of the development and evaluation of hardware systems, as they provide valuable data and insights that can be used to improve the design and performance of these systems.

1. Verification of Kirchhoff's laws

Kirchhoff's laws are two laws that deal with the conservation of charge and energy in electrical circuits. These laws are known as Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL).

- **Kirchhoff's Current Law (KCL):** This law states that the algebraic sum of currents entering and leaving a node in an electrical circuit is zero. In other words, the total current entering a node is equal to the total current leaving the node.
- **Kirchhoff's Voltage Law (KVL):** This law states that the algebraic sum of the voltage drops around any closed loop in an electrical circuit is zero. In other words, the total voltage rise in a closed loop is equal to the total voltage drop in the same loop.

To verify Kirchhoff's laws, one can set up an electrical circuit with known resistances and voltage sources. By measuring the current at different points in the circuit and the voltage drops across different components, one can use KCL and KVL to calculate the expected values and compare them to the measured values. If the calculated and measured values match, it verifies the validity of Kirchhoff's laws.

2. Measurement of power and power factor in a single phase ac series inductive circuit and study improvement of power factor using capacitor

- Power is the rate at which energy is transferred or used. In an AC circuit, power is measured in watts (W).
- Power factor is the ratio of real power to apparent power. It is a measure of how effectively the power is being used in a circuit.
- In a single phase AC series inductive circuit, the power factor is typically less than 1 due to the presence of inductive reactance.
- The power factor can be improved by adding a capacitor in parallel with the inductive load. This is known as power factor correction.
- Power factor correction reduces the reactive power drawn from the power source, which can result in reduced energy costs and improved system efficiency.
- To measure the power and power factor in a single phase AC series inductive circuit, a wattmeter and a power factor meter can be used.
- The wattmeter measures the real power consumed by the circuit, while the power factor meter measures the power factor.
- By comparing the readings from these two meters, the effectiveness of the power factor correction can be determined.

3. Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.

Resonance is a phenomenon that occurs in electric circuits consisting of capacitors and inductors. An RLC series circuit is an electric circuit consisting of a resistor (R), an inductor (L), and a capacitor (C) connected in series. The resonant frequency of an RLC series circuit is the frequency at which the circuit exhibits resonance.

The resonant frequency of an RLC series circuit can be obtained using the formula:

$$f = 1 / (2 * \sqrt{LC})$$

Where: - f is the resonant frequency - L is the inductance of the inductor - C is the capacitance of the capacitor

At the resonant frequency, the impedance of the circuit is at its minimum value, and the current flowing through the circuit is at its maximum value. This means that the circuit is most efficient at transferring energy at the resonant frequency.

In summary, the study of the phenomenon of resonance in an RLC series circuit involves understanding the behavior of the circuit at its resonant frequency, and how to obtain the resonant frequency using the formula provided above. This knowledge can be useful in designing and analyzing electric circuits that involve capacitors and inductors.

4. Connection and measurement of power consumption of a fluorescent lamp (tube light)

A fluorescent lamp, also known as a tube light, is a low-pressure mercury-vapor gas-discharge lamp that uses fluorescence to produce visible light. To measure the power consumption of a fluorescent lamp, the following steps can be taken:

1. **Connect the lamp to a power source:** The fluorescent lamp should be connected to a power source using the appropriate wiring and connections. Ensure that the connections are secure and that the lamp is properly installed.
2. **Measure the current:** Use a clamp meter to measure the current flowing through the lamp. The clamp meter should be placed around one of the wires leading to the lamp, and the current should be measured while the lamp is turned on.
3. **Measure the voltage:** Use a multimeter to measure the voltage across the lamp. The multimeter should be connected to the two wires leading to the lamp, and the voltage should be measured while the lamp is turned on.
4. **Calculate the power consumption:** The power consumption of the lamp can be calculated by multiplying the current and the voltage. The result will be the power consumption of the lamp in watts.

It is important to note that the power consumption of a fluorescent lamp may vary depending on factors such as the age of the lamp, the temperature of the environment, and the voltage of the power source. It is recommended to measure the power consumption of the lamp under normal operating conditions to obtain an accurate result.

5. Measurement of power in 3- phase circuit by two-wattmeter method and determination of its power factor for star as well as delta connected load.

The two-wattmeter method is used to measure the power in a three-phase circuit. This method involves the use of two wattmeters, which are connected to measure the power in two of the three phases. The third phase is not directly measured, but its power can be calculated from the readings of the other two wattmeters.

The power factor of the circuit can also be determined using the two-wattmeter method. The power factor is the ratio of the real power to the apparent power in the circuit. It is a measure of how efficiently the circuit is using the power that is being supplied to it.

The two-wattmeter method can be used to measure the power in both star and delta connected loads. In a star connected load, the wattmeters are connected between the line and the neutral. In a delta connected load, the wattmeters are connected between the lines.

To determine the power factor using the two-wattmeter method, the following steps can be followed:

1. Measure the power in two of the three phases using the two wattmeters.
2. Calculate the power in the third phase using the readings of the two wattmeters.
3. Calculate the total power in the circuit by adding the power in the three phases.
4. Calculate the apparent power in the circuit using the formula: Apparent power = $\sqrt{3} \times \text{Line voltage} \times \text{Line current}$.
5. Calculate the power factor using the formula: Power factor = Real power / Apparent power.

This method is widely used in the industry due to its accuracy and ease of use. It is important to note that the two wattmeters must be calibrated and connected correctly in order to obtain accurate results.

6. Determination of parameters of ac single phase series RLC circuit

An RLC circuit is an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series. The circuit is driven by an alternating current (ac) voltage source. The parameters of an RLC circuit can be determined by measuring the voltage and current at different points in the circuit and using the appropriate equations.

1. **Resistance (R):** The resistance of the circuit can be determined by measuring the voltage across the resistor and the current flowing through it. The resistance is calculated using Ohm's law, which states that the voltage across a resistor is equal to the product of the current flowing through it and its resistance ($V = IR$).
2. **Inductance (L):** The inductance of the circuit can be determined by measuring the voltage across the inductor and the current flowing through it. The inductance is calculated using the formula $V = L * (dI/dt)$, where V is the voltage across the inductor, L is the inductance, and dI/dt is the rate of change of current.
3. **Capacitance (C):** The capacitance of the circuit can be determined by measuring the voltage across the capacitor and the current flowing through it. The capacitance is calculated using the formula $I = C * (dV/dt)$, where I is the current flowing through the capacitor, C is the capacitance, and dV/dt is the rate of change of voltage.
4. **Impedance (Z):** The impedance of the circuit can be determined by measuring the voltage across the circuit and the current flowing through it. The impedance is calculated using Ohm's law, which states that the voltage across a circuit is equal to the product of the current flowing through it and its impedance ($V = IZ$).

5. **Resonant frequency (f):** The resonant frequency of the circuit can be determined by measuring the frequency at which the circuit exhibits maximum impedance. The resonant frequency is calculated using the formula $f = 1 / (2 * \pi * \sqrt{LC})$, where f is the resonant frequency, L is the inductance, and C is the capacitance.

These are the main parameters that can be determined in an ac single phase series RLC circuit. By measuring the voltage and current at different points in the circuit and using the appropriate equations, the values of these parameters can be calculated.

7. Determination of (i) Voltage ratio (ii) polarity and (iii) efficiency by load test of a single phase Transformer

A single-phase transformer is an electrical device that is used to transfer electrical energy from one circuit to another by means of electromagnetic induction. The voltage ratio, polarity, and efficiency of a single-phase transformer can be determined by performing a load test.

- i. **Voltage Ratio:** The voltage ratio of a transformer is the ratio of the secondary voltage to the primary voltage. It can be determined by measuring the primary and secondary voltages of the transformer while it is under load.
- ii. **Polarity:** The polarity of a transformer refers to the direction of the induced voltage in the secondary winding relative to the primary winding. It can be determined by performing a polarity test, which involves connecting the primary and secondary windings in series and measuring the resulting voltage.
- iii. **Efficiency:** The efficiency of a transformer is the ratio of the output power to the input power. It can be determined by measuring the input and output power of the transformer while it is under load.

During a load test, the transformer is connected to a load and the primary and secondary voltages, as well as the input and output power, are measured. These measurements can be used to calculate the voltage ratio, polarity, and efficiency of the transformer. It is important to note that the load test should be performed under the same conditions that the transformer will be operating under in order to obtain accurate results.

8. Determination of efficiency of a dc shunt motor by load test

The efficiency of a DC shunt motor can be determined by performing a load test. The following steps outline the procedure for conducting a load test on a DC shunt motor:

1. Connect the motor to a suitable load, such as a brake or a dynamometer, and adjust the load to the desired value.

2. Measure the input voltage and current to the motor using a voltmeter and an ammeter, respectively.
3. Measure the speed of the motor using a tachometer.
4. Calculate the input power to the motor by multiplying the input voltage and current.
5. Calculate the output power of the motor by multiplying the torque and speed of the motor.
6. Calculate the efficiency of the motor by dividing the output power by the input power and multiplying by 100%.

The efficiency of the motor can be plotted against the load to determine the optimal operating point of the motor. It is important to note that the efficiency of a DC shunt motor will vary with the load, and the motor should be operated at or near its optimal operating point for maximum efficiency.

9. To study running and speed reversal of a three phase induction motor and record speed in both directions.

- A three-phase induction motor is an AC motor that operates on the principle of electromagnetic induction.
- The motor consists of a stator, which is a stationary part, and a rotor, which is a rotating part.
- The stator has a three-phase winding, which is connected to a three-phase AC supply.
- When the three-phase AC supply is applied to the stator winding, a rotating magnetic field is produced.
- The rotating magnetic field induces an electromotive force (EMF) in the rotor, which causes the rotor to rotate.
- The speed of the rotor is determined by the frequency of the AC supply and the number of poles in the stator winding.
- The direction of rotation of the rotor can be reversed by changing the sequence of the three-phase AC supply to the stator winding.
- To record the speed of the motor in both directions, a tachometer can be used.
- A tachometer is an instrument that measures the rotational speed of a shaft or disk.
- The tachometer can be attached to the shaft of the motor, and the speed can be read from the display of the tachometer.
- By reversing the sequence of the three-phase AC supply to the stator winding, the direction of rotation of the rotor can be reversed, and the speed in the reverse direction can be recorded using the tachometer.

10. Demonstration of cut-out sections of machines: dc machine, three phase induction machine, single-phase induction machine and synchronous machine.

- **DC Machine:** A DC machine is an electromechanical device that con-

verts electrical energy into mechanical energy or vice versa. The cut-out section of a DC machine shows the armature, commutator, brushes, and field winding.

- **Three Phase Induction Machine:** A three-phase induction machine is an AC motor that converts electrical energy into mechanical energy. The cut-out section of a three-phase induction machine shows the stator, rotor, and windings.
- **Single-Phase Induction Machine:** A single-phase induction machine is an AC motor that converts electrical energy into mechanical energy. The cut-out section of a single-phase induction machine shows the stator, rotor, and windings.
- **Synchronous Machine:** A synchronous machine is an AC machine that converts electrical energy into mechanical energy or vice versa. The cut-out section of a synchronous machine shows the stator, rotor, field winding, and damper winding.

These cut-out sections provide a visual representation of the internal components and workings of these machines, allowing for a better understanding of their operation and maintenance.

(B) Experiments available on virtual lab

There are many virtual labs available online that offer a wide range of experiments for students to perform. Some of the virtual labs available are:

1. **Virtual Chemistry Experiments** - A collection of interactive web-based chemistry tutorials .
2. **Mixed Reception** - Solve a mysterious death by gathering clues, interviewing suspects, and analyzing evidence .
3. **The Virtual Lab** - An online simulation of a chemistry lab designed to help students link chemical computations with authentic laboratory chemistry .
4. **PraxiLabs** - Offers virtual experiments that are adaptive to any learning style, with progress tracking, walkthrough videos, lab manuals, enrichment documents, immediate feedback, MCQ quizzes, and much more .
5. **VRLab Academy** - Offers more than 200 science experiments available on VR and PC, with virtual reality technology to increase the ability to learn through virtual memory .
6. **CK-12 Chemistry Simulations** - Nearly two dozen simulations cover topics like average atomic mass, solubility with rock candy, and freezing point depression with road salt .
7. **Mystery Science** - Offers a wealth of virtual science labs, activities, and experiments for students in elementary school, covering topics from weather to astronomy, humans to animals .

8. **Molecular Workbench** - Offers hundreds of virtual science labs for physics, biology, chemistry, and more .
9. **National Science Digital Library: ChemCollective Virtual Labs** - Take a deep dive into chemistry with these virtual labs .
10. **PBS: Nova Labs** - A terrific platform for teens and older learners to explore and examine cutting-edge issues .

These are just a few examples of the many virtual labs available online that offer a wide range of experiments for students to perform. These virtual labs provide an excellent opportunity for students to learn and explore science in a safe and interactive environment.

1. Kirchhoff's laws

Kirchhoff's laws are two laws that deal with the conservation of charge and energy in electrical circuits. These laws are named after the German physicist Gustav Kirchhoff.

1. Kirchhoff's Current Law (KCL): This law states that the algebraic sum of currents entering a node (or a closed boundary) is zero. In other words, the total current entering a node is equal to the total current leaving the node. This law is based on the principle of conservation of electric charge.
2. Kirchhoff's Voltage Law (KVL): This law states that the algebraic sum of the potential differences (voltages) around any closed loop is zero. In other words, the total voltage rise in a closed loop is equal to the total voltage drop. This law is based on the principle of conservation of energy.

These laws are widely used in circuit analysis to determine the current, voltage, and power in a circuit. They are fundamental principles that must be satisfied in any circuit, regardless of its complexity.

Virtual Lab

Virtual Lab at Amrita uses state-of-the-art computer simulation technology to create real-world environments and problem handling capabilities. It is required to bridge the gap between institutions (or industries) that retain the physical laboratory and distantly placed economically challenged educational institutions in areas such as Physics, Chemistry, and Biotechnology.

Virtual labs are media-rich online learning environments, which allow users to conduct physical laboratory experiments in a computer-simulated environment. This innovative new educational technique is enabling experimentation beyond the boundaries of traditional laboratories.

The Virtual Lab at Amrita Vishwa Vidyapeetham is called VALUE (Virtual Amrita Laboratories Universalizing Education). It provides an interactive environment for conducting experiments and performing activities.

One of the experiments available in the Virtual Lab is Kirchhoff's Laws. This experiment allows users to learn about the principle known as Kirchhoff's Voltage Law, which states that "The algebraic sum of all voltages in a loop must equal zero."

2. Thevenin Theorem

Thevenin's Theorem is a technique used to analyze complex electrical circuits by simplifying them into an equivalent circuit composed of a single voltage source, a single resistor, and a load. This theorem is named after Léon Charles Thévenin, a French telegraph engineer who first stated the theorem in 1883.

The steps to apply Thevenin's Theorem are as follows:

1. Identify the load resistor in the circuit, which is the resistor across which the voltage or current is to be determined.
2. Remove the load resistor from the circuit.
3. Calculate the open-circuit voltage across the terminals where the load resistor was connected. This voltage is known as the Thevenin voltage (V_{th}).
4. Calculate the equivalent resistance of the circuit as seen from the open terminals, with all voltage sources replaced by short circuits and all current sources replaced by open circuits. This resistance is known as the Thevenin resistance (R_{th}).
5. Replace the original circuit with an equivalent circuit composed of a voltage source with a value equal to the Thevenin voltage (V_{th}) in series with a resistor with a value equal to the Thevenin resistance (R_{th}).
6. Reconnect the load resistor to the equivalent circuit and solve for the desired voltage or current.

Thevenin's Theorem is a powerful tool for analyzing complex circuits and can greatly simplify the process of circuit analysis. It is widely used in electrical engineering and related fields.

Virtual Lab

Virtual Lab is an initiative by Amrita University, Dayalbagh University, NIT Karnataka, and COE Pune. The main objective of Virtual Labs is to provide remote access to labs in various disciplines of Science and Engineering. These Virtual Labs cater to students at the undergraduate level, postgraduate level, and research scholars .

Amrita Virtual Biotechnology laboratory is a new kind of experimental science that is being established as a virtual simulation-based laboratory. These labs focus on helping students retain the real feel of a laboratory while conducting the experiment from an internet-enabled computer terminal, much in the same way as is done in a real lab .

The VALUE Initiative is a project dedicated to developing these virtual labs in various science and engineering disciplines. Virtual Lab at Amrita uses state-of-the-art computer simulation technology to create real-world environments and problem-handling capabilities. This is required to bridge the gap between institutions (or industries) that retain the physical laboratory and distantly placed economically challenged educational institutions in areas such as Physics .

Optics Virtual Lab is one of the labs available under the Physical Sciences category. Optics is the study of the behavior and properties of light, including its interactions with matter and the construction of instruments that use or detect it. Optics usually describes the behavior of visible, ultraviolet, and infrared light .

The Online Labs is based on the idea that lab experiments can be taught using the Internet, more efficiently, and less expensively. The labs can also be made available to students with no access to physical labs or where equipment is not available owing to being scarce or costly .

3. RLC Series Resonance

An RLC series circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series. At a certain frequency, called the resonant frequency, the impedance of the circuit is at its minimum and the current is at its maximum.

- The resonant frequency of an RLC series circuit is given by the formula: $f = 1 / (2 * \pi * \sqrt{L * C})$.
- At resonance, the inductive reactance and the capacitive reactance are equal in magnitude and opposite in sign, resulting in a net reactance of zero.
- The impedance of the circuit at resonance is equal to the resistance of the resistor.
- The voltage across the inductor and the capacitor are equal in magnitude and opposite in phase.
- The current in the circuit is in phase with the voltage across the resistor.
- The power factor of the circuit at resonance is 1, meaning that all the power supplied by the source is dissipated in the resistor.
- The quality factor (Q) of the circuit is given by the formula: $Q = (2 * \pi * f * L) / R$, where f is the resonant frequency.
- The bandwidth of the circuit is given by the formula: $BW = f / Q$, where f is the resonant frequency and Q is the quality factor.

Virtual Lab

Virtual Labs are online platforms that provide students with an interactive space to engage with their subject matter. They can be used for pre-lab preparation, online lab courses, post-lab review, make-up labs, or used with a textbook to bridge the gap between lab and lecture courses . Virtual Labs are available for

a wide range of disciplines, including Anatomy & Physiology, Biology (Majors and Non-majors), Botany, Environmental Science, Genetics, Human Biology, Medical Assisting, Microbiology, and Nutrition .

Some benefits of Virtual Labs include: - Providing remote-access to simulation-based Labs in various disciplines of Science and Engineering . - Enthusing students to conduct experiments by arousing their curiosity . - Helping students learn basic and advanced concepts through remote experimentation .

Virtual Labs can be a valuable tool for students and educators alike, offering a robust online professional development system . They are available as stand-alone products or can be assignable within certain educational platforms . There are also many free virtual labs available, designed for high school and college students, covering a wide range of topics .

4. Measurement of power in 3- phase circuit by two wattmeter method and determination of its power factor for star as well as delta connected load.

The two wattmeter method is a technique used to measure the power in a three-phase circuit. This method involves the use of two wattmeters, which are connected to the circuit in such a way that one wattmeter measures the power in one of the phases, while the other wattmeter measures the power in the other two phases.

The power factor of the circuit can be determined by calculating the ratio of the active power to the apparent power. This can be done by measuring the power in each of the phases using the two wattmeter method, and then using these measurements to calculate the power factor.

For a star-connected load, the power factor can be determined by dividing the sum of the two wattmeter readings by the square root of three times the line voltage multiplied by the line current.

For a delta-connected load, the power factor can be determined by dividing the difference of the two wattmeter readings by the line voltage multiplied by the line current.

In summary, the two wattmeter method is a useful technique for measuring the power in a three-phase circuit and determining its power factor, for both star and delta connected loads. It involves the use of two wattmeters, which are connected to the circuit in a specific way, and the power factor can be calculated using the measurements obtained from these wattmeters.

Virtual Lab

Virtual Lab is an online platform that provides students with the opportunity to perform experiments and learn about science and engineering subjects. The Virtual Power Laboratory-D.E.I. is one such lab that allows students to study the

synchronization of an alternator with an infinite bus bar, determine the direct axis reactance (X_d) and quadrature axis reactance (X_q) of a synchronous machine, and learn about measuring devices such as ammeters, voltmeters, wattmeters, and power factor meters .

The goal of the Virtual Power Lab is to develop virtual machines and power laboratory to facilitate the students to work with their own convenience and easily understand the practical concepts. It also helps in generating interest in the student community to get better insight about the subjects related to machines and power systems .

Virtual Labs are intended to augment the learning of science and engineering subjects through performing experiments. The experiments are designed either as simulations or as remote triggered. A remote triggered lab allows a user to connect to real equipment using a web browser .

5. Determination of parameters of ac single phase series RLC circuit

An RLC circuit is an electrical circuit consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series. The behavior of an RLC circuit is determined by the values of the resistance, inductance, and capacitance, as well as the frequency of the applied alternating current (ac).

To determine the parameters of an ac single phase series RLC circuit, the following steps can be followed:

1. Measure the resistance (R) of the resistor using an ohmmeter.
2. Measure the inductance (L) of the inductor using an inductance meter.
3. Measure the capacitance (C) of the capacitor using a capacitance meter.
4. Calculate the resonant frequency (f) of the circuit using the formula $f = \frac{1}{2\pi\sqrt{LC}}$.
5. Calculate the impedance (Z) of the circuit at the resonant frequency using the formula $Z = R + j(2\pi fL - \frac{1}{2\pi fC})$.
6. Calculate the current (I) flowing through the circuit at the resonant frequency using the formula $I = V / Z$, where V is the applied voltage.
7. Calculate the voltage drop across each component using Ohm's law ($V = I * R$ for the resistor, $V = j * I * 2\pi * f * L$ for the inductor, and $V = -j * I / (2\pi * f * C)$ for the capacitor).

These calculations will allow you to determine the parameters of an ac single phase series RLC circuit. It is important to note that the values of the resistance, inductance, and capacitance may vary with frequency, so it is important to measure them at the frequency of interest. Additionally, the presence of parasitic elements, such as stray capacitance and inductance, can affect the behavior of the circuit, so it is important to take these into account when performing the calculations.

Virtual Lab

Virtual labs are interactive tools that offer a space for students to engage with their subject matter. They can be used for pre-lab preparation, online lab courses, post-lab review, make-up labs, or used with a textbook to bridge the gap between lab and lecture courses . Virtual labs are available in various disciplines such as Anatomy & Physiology, Biology (Majors and Non-majors), Botany, Environmental Science, Genetics, Human Biology, Medical Assisting, Microbiology, and Nutrition .

Virtual labs can be used to provide remote-access to simulation-based Labs in various disciplines of Science and Engineering. They can help students learn basic and advanced concepts through remote experimentation . Virtual labs can also be used to enthuse students to conduct experiments by arousing their curiosity .

Virtual labs can be assigned within educational platforms such as McGraw Hill ALEKS and Connect, and are also available as stand-alone products . They can be used in the remote learning context and are available for both science and non-science disciplines .

6. To observe the B-H loop of a ferromagnetic material in CRO

- The B-H loop of a ferromagnetic material is a graphical representation of the relationship between the magnetic field strength (H) and the magnetic flux density (B) of the material.
- The B-H loop is also known as the hysteresis loop.
- The loop is obtained by subjecting the material to an alternating magnetic field and measuring the resulting changes in the magnetic flux density.
- The shape of the B-H loop is characteristic of the ferromagnetic material and provides information about its magnetic properties.
- A Cathode Ray Oscilloscope (CRO) can be used to observe the B-H loop of a ferromagnetic material.
- To do this, the material is placed inside a solenoid and subjected to an alternating current. The changes in the magnetic flux density are measured using a search coil and displayed on the CRO.
- The resulting trace on the CRO screen is the B-H loop of the ferromagnetic material.
- The area enclosed by the B-H loop is a measure of the energy lost in the material due to hysteresis.
- The shape of the B-H loop can be affected by factors such as the frequency of the alternating current, the temperature of the material, and the presence of impurities or defects in the material.

Virtual Lab

Virtual labs are online tools that offer a space for students to engage with their subject matter interactively. They can be used for pre-lab preparation,

online lab courses, post-lab review, make-up labs, or used with a textbook to bridge the gap between lab and lecture courses. Virtual labs are available for a wide range of disciplines, including Anatomy & Physiology, Biology (Majors and Non-majors), Botany, Environmental Science, Genetics, Human Biology, Medical Assisting, Microbiology, and Nutrition.

Virtual labs provide remote access to simulation-based labs in various disciplines of Science and Engineering. They are designed to enthuse students to conduct experiments by arousing their curiosity, helping them learn basic and advanced concepts through remote experimentation. Some virtual labs include built-in quizzes, notebooks, and references to enhance the learning experience.

Virtual labs are a valuable resource for remote learning, allowing students to engage in interactive, hands-on learning experiences from the comfort of their own homes. They are available from a variety of sources, including educational institutions, publishers, and independent organizations.

7. Determination of the efficiency of a dc motor by loss summation method (Swinburne's test)

Swinburne's test is an indirect method of testing DC machines, named after Sir James Swinburne. It is the simplest method of testing shunt and compound DC machines which have constant field flux. In this method, the DC machine (generator or motor) is run as a motor at no-load and losses of the machine are determined. Once the losses of the machine are known, its efficiency at any desired load can be determined in advance.

The Swinburne's test on DC shunt motor is applicable to find the losses in the machine with no-load power. The losses in the motors are armature copper losses, iron losses in the core, friction losses, and winding losses. These losses are calculated separately and efficiency can be pre-determined.

Course Outcomes:

1. Understand the fundamental concepts and principles of the subject.
2. Develop the ability to apply the knowledge and skills acquired in the course to solve problems.
3. Enhance critical thinking and analytical skills.
4. Improve communication and teamwork skills.
5. Gain an appreciation for the importance of lifelong learning and professional development.
6. Develop an understanding of ethical and social responsibilities in the field.

Course Outcome (CO) Bloom's Level

Course outcomes (COs) are statements that describe the knowledge, skills, and abilities that students should possess and can demonstrate upon completion of

a course. COs are typically written using action verbs that align with Bloom's Taxonomy, a framework for categorizing educational goals and objectives.

Bloom's Taxonomy is a hierarchical model that classifies learning objectives into different levels of complexity and specificity. The levels, in ascending order, are: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.

- **Remembering:** The ability to recall or retrieve previously learned information.
- **Understanding:** The ability to comprehend the meaning of instructional messages, including oral, written, and graphic communication.
- **Applying:** The ability to use learned material in new and concrete situations.
- **Analyzing:** The ability to break down material into its component parts so that its organizational structure may be understood.
- **Evaluating:** The ability to judge the value of material for a given purpose.
- **Creating:** The ability to put parts together to form a new whole.

When writing COs, it is important to ensure that they align with the appropriate level of Bloom's Taxonomy. This helps to ensure that the course is designed to facilitate the achievement of the desired learning outcomes. For example, a CO that states "Students will be able to analyze the impact of historical events on contemporary society" aligns with the Analyzing level of Bloom's Taxonomy.

In summary, COs are statements that describe the desired learning outcomes of a course, and Bloom's Taxonomy provides a framework for categorizing these outcomes into different levels of complexity and specificity. By aligning COs with the appropriate level of Bloom's Taxonomy, instructors can design courses that effectively facilitate student learning and achievement.

At the end of this course, the students should be able to:

1. Understand the fundamental concepts and principles of the subject.
2. Apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
3. Communicate effectively using the language and terminology of the subject.
4. Work collaboratively with others to achieve common goals.
5. Demonstrate critical thinking and analytical skills in evaluating information and arguments.
6. Use appropriate technology and tools to enhance learning and productivity.
7. Develop a lifelong interest in the subject and a commitment to continuous learning.

CO 1 Conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits. K3

- KVL (Kirchhoff's Voltage Law) states that the sum of the voltage drops around a closed loop in a circuit is equal to the voltage supplied to the loop.
- KCL (Kirchhoff's Current Law) states that the sum of the currents entering a node is equal to the sum of the currents leaving the node.
- Network theorems such as Thevenin's theorem, Norton's theorem, and Superposition theorem can be used to analyze and simplify complex DC electrical circuits.
- To conduct experiments illustrating the application of KVL/KCL and network theorems to DC electrical circuits, one can set up a circuit with known voltage sources and resistors and measure the voltage drops and currents using a multimeter.
- By applying KVL and KCL to the circuit, one can verify that the measured values match the calculated values.
- Similarly, by applying network theorems, one can simplify the circuit and verify that the simplified circuit behaves in the same way as the original circuit.
- These experiments can help to deepen the understanding of the fundamental principles of DC electrical circuits and the application of KVL/KCL and network theorems.

CO 2 Demonstrate the behavior of AC circuits connected to single phase AC supply and measure power in single phase as well as three phase electrical circuits. K4

- AC circuits are electrical circuits that are powered by an alternating current (AC) source.
- A single-phase AC supply is an electrical power supply that delivers power through a single conductor.
- The behavior of an AC circuit connected to a single-phase AC supply can be demonstrated by measuring the voltage, current, and power in the circuit.
- Power in a single-phase AC circuit can be measured using a wattmeter, which measures the product of the voltage and current in the circuit.
- A three-phase electrical circuit is a type of polyphase system that uses three conductors to deliver power.
- Power in a three-phase electrical circuit can be measured using a three-phase wattmeter, which measures the power in each of the three phases and sums them to give the total power in the circuit.
- Understanding the behavior of AC circuits and the ability to measure power in single-phase and three-phase electrical circuits is important for the design and analysis of electrical systems.

CO 3 Perform experiment illustrating BH curve of magnetic materials. K3

The BH curve of magnetic materials is a graphical representation of the relationship between magnetic field strength (H) and magnetic flux density (B) within a magnetic material. To perform an experiment illustrating the BH curve of magnetic materials, the following steps can be followed:

1. **Select a magnetic material:** Choose a magnetic material for which you want to determine the BH curve. Common magnetic materials include iron, cobalt, nickel, and their alloys.
2. **Prepare the sample:** Prepare the sample of the magnetic material in the form of a ring or toroid. This shape is preferred as it minimizes the leakage of the magnetic field.
3. **Set up the experiment:** Set up the experiment by winding a primary coil of wire around the magnetic material sample. Connect the primary coil to an AC power source. Also, wind a secondary coil of wire around the sample and connect it to an oscilloscope or a flux meter.
4. **Apply a magnetic field:** Apply a magnetic field to the magnetic material sample by passing an alternating current through the primary coil. The magnetic field strength (H) can be calculated using the formula $H = NI/L$, where N is the number of turns in the primary coil, I is the current passing through the primary coil, and L is the length of the magnetic path.
5. **Measure the magnetic flux density:** Measure the magnetic flux density (B) of the magnetic material sample using the secondary coil. The voltage induced in the secondary coil is proportional to the rate of change of the magnetic flux. The magnetic flux density can be calculated using the formula $B = \Phi/A$, where Φ is the magnetic flux and A is the cross-sectional area of the magnetic material sample.
6. **Plot the BH curve:** Plot the BH curve by graphing the magnetic field strength (H) on the x-axis and the magnetic flux density (B) on the y-axis. The shape of the BH curve will depend on the magnetic properties of the material being tested.

CO 4 Calculate efficiency of a single phase transformer and DC machine. K4

Efficiency is defined as the ratio of output power to input power. It is a measure of how well a device converts the power it receives into useful work.

To calculate the efficiency of a single phase transformer, the following steps can be taken:

1. Measure the input voltage and current to the transformer.
2. Measure the output voltage and current from the transformer.
3. Calculate the input power by multiplying the input voltage by the input current.

4. Calculate the output power by multiplying the output voltage by the output current.
5. Divide the output power by the input power to obtain the efficiency of the transformer.

Similarly, to calculate the efficiency of a DC machine, the following steps can be taken:

1. Measure the input voltage and current to the DC machine.
2. Measure the output power from the DC machine.
3. Calculate the input power by multiplying the input voltage by the input current.
4. Divide the output power by the input power to obtain the efficiency of the DC machine.

It is important to note that the efficiency of a transformer or a DC machine can vary depending on the load and operating conditions. Therefore, it is recommended to perform the efficiency calculation under the specific operating conditions for which the device is intended to be used.

CO 5 Perform experiments on speed measurement and reversal of direction of three phase induction motor and Identify the type of DC and AC machines based on their construction. K4

1. **Speed measurement of three phase induction motor:** The speed of a three-phase induction motor can be measured using a tachometer, stroboscope, or by counting the number of revolutions per minute (RPM) of the motor shaft. The synchronous speed of the motor can be calculated using the formula $N_s = 120f/P$, where N_s is the synchronous speed, f is the frequency of the supply, and P is the number of poles.
2. **Reversal of direction of three phase induction motor:** The direction of rotation of a three-phase induction motor can be reversed by interchanging any two of the three supply lines. This changes the direction of the rotating magnetic field, causing the rotor to rotate in the opposite direction.
3. **Identification of DC and AC machines based on their construction:** DC machines can be identified by the presence of a commutator, which is used to convert the alternating current (AC) produced in the armature windings into direct current (DC). AC machines, on the other hand, do not have a commutator and rely on the alternating nature of the supply to produce a rotating magnetic field. The construction of the rotor is also different, with DC machines having a wound rotor and AC machines having either a wound rotor or a squirrel-cage rotor.

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

These are the six levels of Bloom's Taxonomy, a framework for categorizing educational goals and objectives. Each level represents a different type of cognitive skill that students should develop.

1. **K1 – Remember:** This level involves recalling information from memory. Examples of tasks at this level include defining terms, listing facts, and identifying concepts.
2. **K2 – Understand:** This level involves comprehending the meaning of information. Examples of tasks at this level include explaining concepts, summarizing information, and interpreting data.
3. **K3 – Apply:** This level involves using information in a new situation. Examples of tasks at this level include solving problems, applying theories, and demonstrating techniques.
4. **K4 – Analyze:** This level involves breaking down information into parts and examining their relationships. Examples of tasks at this level include comparing and contrasting, categorizing, and finding patterns.
5. **K5 – Evaluate:** This level involves making judgments based on criteria and standards. Examples of tasks at this level include evaluating the credibility of sources, assessing the quality of arguments, and making decisions.
6. **K6 – Create:** This level involves combining information to form a new whole. Examples of tasks at this level include designing, constructing, and planning.

These levels can be used to design educational activities and assessments that target different types of cognitive skills. It is important for students to develop skills at all levels of the taxonomy in order to become well-rounded learners.